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COA

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CAPSTONE ASSIGNMENT

Case Study 1 : Smart Traffic Control System

1.) Role of Basic Functional Units

- Input Unit : Collects data from traffic sensors (vehicles, signals, cameras)
- Memory Unit : Stores sensor data, control algorithms, and traffic rules.
- ALU (Arithmetic Logic Unit) : Processes data and makes decisions (e.g., signal timing).
- Control Unit : Coordinates all operations and manages real-time responses.
- Output Unit : Sends commands to traffic lights and display systems.

2.) RISC vs CISC architecture

A RISC (Reduced Instruction Set Computer) architecture is more suitable because :

- Faster execution due to simple instructions
- Better for real-time processing
- Lower power consumption
- Easier pipelining compared to CISC

3.) Instruction Throughput (IPC) and Performance

- Higher IPC (Instructions Per Cycle) improves system responsiveness.
- Faster processing ensures quick reaction to traffic changes, reducing congestion.
- Good performance is critical for real-time decision-making

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4) Processor Type Selection

A pipelined processor is the best choice because :

- Executes multiple instructions simultaneously.
- Improves efficiency and speed.
- Suitable for continuous data processing in real - time systems.

5) Pipeline Hazards and Solutions

- Data Hazards : Occur due to dependency between instructions
  → Solution : Forwarding , stalling

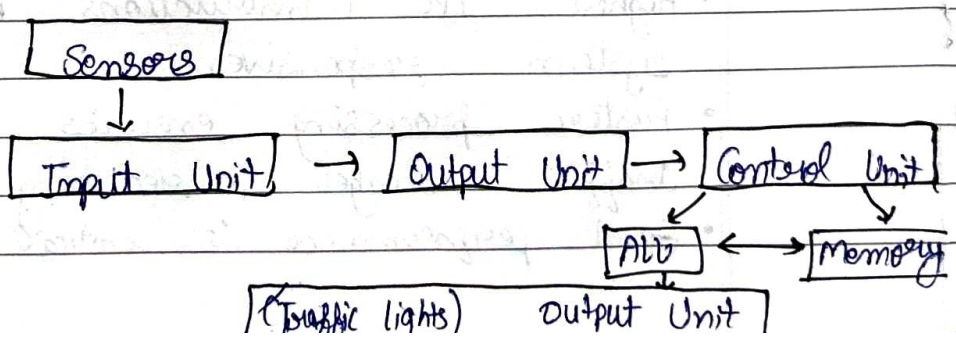
- Control Hazards : Due to branch instructions.
  → Solution : Branch prediction

- Structural Hazards : Resource conflicts
  → Solution : Better hardware design or resource duplication

⇒ Conclusion

A RISC - based pipelined processor with high IPC is ideal for a smart traffic control system due to its speed , efficiency , and ability to handle real - time operations reliably.

Flow Diagram



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Case Study 2 : Storage Planning for a Video Surveillance System

1) Role of Secondary Storage
Secondary storage is used to store large volumes of video data continuously. It provides high capacity, long-term storage, and ensures recorded footage can be retrieved later for security analysis.

2.) Magnetic Disks vs Flash Memory
- Magnetic Disks (HDD) :
  - → High storage capacity
  - → Low cost per GB
  - → Slower access speed

- Flash Memory (SSD) :
  - → Faster data access
  - → More reliable (no moving parts)
  - → Higher cost and lower capacity compared to HDD.

3) Suitable Usage
- HDDs : Best for long-term video storage and archives.

- SSDs : Best for real-time recording, buffering and quick access data.

4.) Memory Hierarchy and Retrieval Speed
Memory hierarchy (Cache → RAM → SSD → HDD) affects speed :

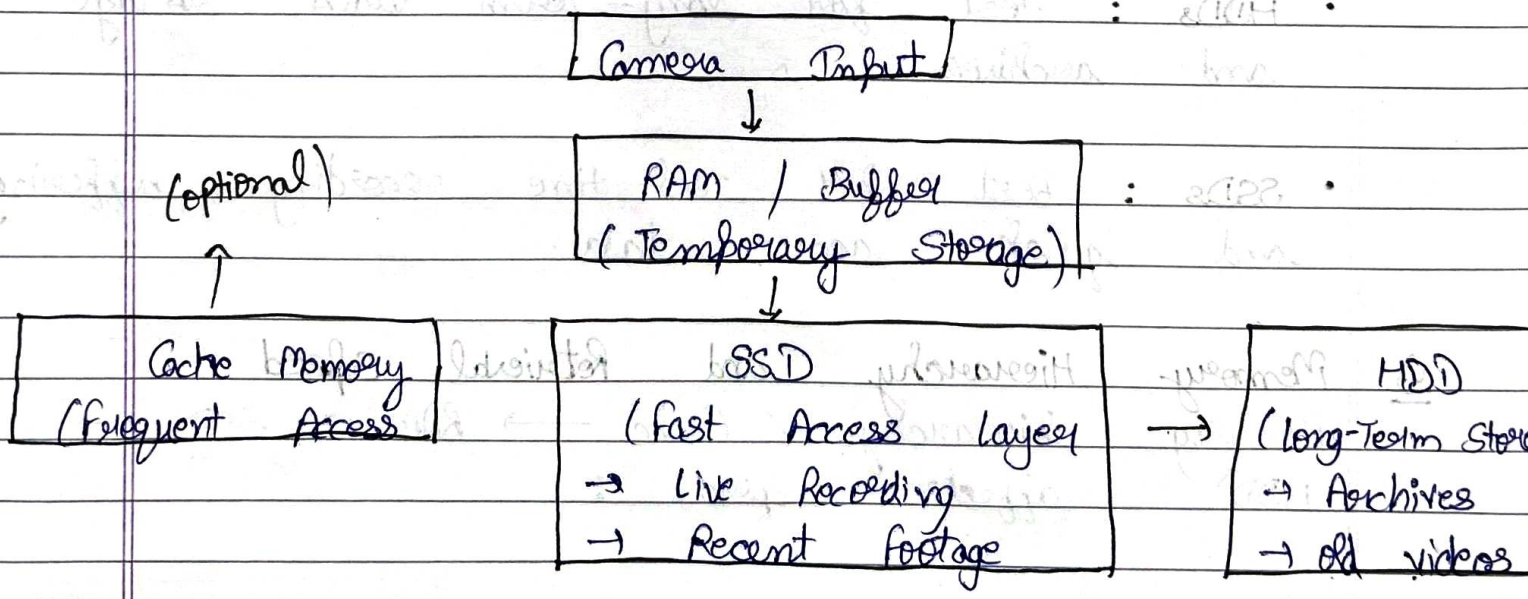
- Data stored in faster memory (cache / SSD) is accessed quickly).
- Data in HDD takes more time.
- Frequently accessed video should be kept in faster storage to improve performance.

5) Optimized Architecture

Use a hybrid storage system :

- SSD layer : For live recording and recent footage.
- HDD layer : For long-term storage.
- Optionally include cache memory for frequently accessed clips.

This balances cost (HDD), speed (SSD), and capacity (HDD + SSD combination).

⇒ Conclusion :

A hybrid architecture using SSDs and HDDs is ideal, ensuring fast access, large storage capacity, and cost efficiency for a video surveillance system.